SIMBA: Mathematical Foundations and Implementation Contract

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15 October 2026

Abstract

SIMBA is a bundle-completion procedure combining a pairwise collaborative backbone, historical joint-incidence surplus, redundancy-weighted quadratic seed concordance, and exposure-modulated Gaussian shrinkage. This document specifies the operators implemented in the standalone repository and derives their mathematical properties. It proves a separation between pairwise projection and triple incidence, characterizes concordance through elementary symmetric polynomials and Rao diversity, derives the normal–normal posterior, and states the exact conditions under which inverse-frequency variance components are identifiable. It also distinguishes algebraic identities, model-conditional statistical results, implementation policies, and empirical retrieval claims. The constituent results on hypergraphs, diversity, and Gaussian conjugacy are established mathematics; their use here does not constitute a claim to have invented those primitives.

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1 Purpose, scope, and types of claims

This is the detailed theoretical companion to `simba/operators.py`. The shorter paper methodology is in `paper/method.tex`. The discussion applies to the current operators, rather than to every historical SIMBA experiment. Dataset-specific evaluators add feature channels and select coefficients; they do not all instantiate an identical scoring formula.

Four kinds of statements must be kept separate:

1. **Algebraic facts:** identities and operator properties that hold for the specified inputs, independent of a data-generating model.
2. **Model-conditional results:** posterior or moment-estimation statements that require explicit distributional or conditional-moment assumptions.
3. **Software policies:** choices such as positive variance floors, clipping, validation grids, and disabling unidentified shrinkage.
4. **Empirical claims:** improvements in Recall@20 or NDCG@20, which require an evaluation protocol and measured evidence.

For example, joint incidence can contain information absent from a pairwise projection. This does not prove that a particular incidence bonus improves ranking. Similarly, a posterior mean minimizes squared-error Bayes risk under its assumed model, but that result is not a Recall@20 guarantee.

2 Task, notation, and information boundary

2.1 Bundle completion

Let $\mathcal{I} = \{1, \dots, n\}$ be a fixed item catalogue and $\mathcal{B}_{\text{tr}} = \{b_1, \dots, b_m\}$ the training bundles. Each bundle is a set; repeated item identifiers within a bundle are removed by the graph constructor. A query consists of an observed seed set $S \subset \mathcal{I}$. The candidate domain is $\mathcal{I} \setminus S$. At evaluation time, a disjoint held-out set H supplies the relevant completion items.

Define the binary incidence matrix and degrees by

$$B_{bi} = \mathbf{1}\{i \in b\}, \qquad f_i = \sum_b B_{bi}, \qquad d_b = \sum_i B_{bi}. \tag{1}$$

The seed indicator is $q_S \in \{0,1\}^n$; a batch of Q queries forms $J = [q_{S_1}, \dots, q_{S_Q}] \in \{0,1\}^{n \times Q}$. Unless explicitly stated otherwise, all graph results below use binary B. The optional weighted constructor is treated separately in Section 13.2.

Symbol	Meaning
$a_i = (f_i + 5)^{-\alpha}$	Seed exposure attenuation
$v_b = (d_b + 1)^{-\nu}$	Bundle-volume attenuation
$r_b = (d_b + 1)^{-1}$	Smoothing bundle normalization
$\beta \geq 0$	Joint-incidence coefficient
$\tau > 0$	Concordance denominator regularizer
$w_{cs} \geq 0$	Candidate support from seed s
G_S, K_S	Seed similarity and concordance weighting matrices
Y_{cS}, μ_{cS}	Fused observation score and scaled prior mean
σ_0^2, σ_ϵ^2	Prior and exposure-noise variance components
$\kappa = \sigma_\epsilon^2 / \sigma_0^2$	Exposure shrinkage scale

The graph defaults are $\alpha = \nu = 1/2$ and $\beta = 1.5$. The code calls ν `gamma_vol`; it is unrelated to the concordance kernel parameter γ used later.

2.2 What each split may supply

Training bundles determine collaborative incidence and exposure. Catalogue metadata and frozen content features describe items, including items unseen in training bundles. Validation queries determine score scaling, variance components, and any selected fusion or shrinkage parameters. Test targets determine evaluation metrics only. Catalogue-level text preprocessing may use all item descriptions; that is a transductive catalogue assumption, not evidence that the corresponding held-out bundle memberships are known.

The benchmark loaders also have dataset-specific auxiliary interactions, filters, and query caps. Those choices are documented in `docs/REPRODUCIBILITY.md`. A mathematical property of the shared operator does not certify a loader's experimental protocol.

3 Pairwise induction and propagation

3.1 Additive pairwise support

Set $D_a = \mathrm{diag}(a)$ and $D_v = \mathrm{diag}(v)$. Before diagonal removal, the pairwise score matrix is

$$W^{\mathrm{full}} = B^\top D_v B D_a, \qquad W = W^{\mathrm{full}} - \mathrm{diag}(\mathrm{diag}(W^{\mathrm{full}})). \tag{2}$$

Thus $W_{cs} = a_s \sum_{b:\, c,s\in b} v_b$ for $c \neq s$. The attenuation is on the seed side, so W need not be symmetric. The additive candidate score is

$$A_c^{\mathrm{pair}}(S) = \sum_{s\in S} W_{cs}. \tag{3}$$

`score_pairwise_and_diff2` computes $W^{\mathrm{full}}J$ in factored form. For $c \notin S$, this is exactly (3). Seed rows can retain self-contributions in the intermediate array; the evaluators exclude seed items from the final candidate ranking. Those intermediate values can still enter propagated channels.

3.2 Symmetric smoothing is not a Markov transition

The binary-incidence implementation of `propagate` applies

$$P_{\mathrm{sym}} = D_a B^\top D_r B D_a - D_h, \quad h_i = a_i^2 \sum_b r_b B_{bi}, \quad D_r = \mathrm{diag}(r). \tag{4}$$

Its diagonal is zero; for $i \neq j$, $(P_{\mathrm{sym}})_{ij} = a_i a_j \sum_b r_b B_{bi} B_{bj} \geq 0$. The matrix is symmetric, but its rows are not generally normalized. Removing the diagonal also need not preserve positive semidefiniteness. Consequently, symmetry alone gives neither a probability transition nor a contraction in a chosen norm. "Diffusion" here denotes graph smoothing. The function returns $P_{\mathrm{sym}}X$; naming a channel "diff2" does not mean that this function itself computes $P_{\mathrm{sym}}^2 X$.

3.3 The separately exposed random-walk operator

For $f_i > 0$ and $d_b > 0$, let

$$T_0 = D_{1/f} B^\top D_{1/d} B, \qquad \bar{T} = T_0 - \mathrm{diag}(\mathrm{diag}(T_0)), \qquad \rho_i = \sum_j \bar{T}_{ij}. \tag{5}$$

Zero-degree reciprocals are defined to be zero. Define

$$(P_{\mathrm{rw}})_{ij} = \begin{cases} \bar{T}_{ij}/\rho_i, & \rho_i > 0, \\ 0, & \rho_i = 0. \end{cases} \tag{6}$$

Proposition 3.1 (Active-row stochasticity)**.** *P_{rw} has nonnegative entries and zero diagonal. Every row with $\rho_i > 0$ sums to one; every other row is zero. On active rows, $(P_{\mathrm{rw}}x)_i$ is a convex combination of neighbouring values.*

Proof. All factors of T_0 are nonnegative. Diagonal deletion preserves off-diagonal nonnegativity. Dividing a positive row sum by itself gives one, and a zero row sum in a nonnegative matrix implies an all-zero row. The convex combination statement follows directly from these weights. □

A warm item occurring only in singleton bundles can also have a zero row. Warm exposure alone is therefore insufficient for the active-row condition. The evaluators do not all use P_{rw}; some use (4) or a graph mixture specific to their dataset. These operators are related to established hypergraph constructions [1], but the precise normalizations above specify the implementation contract.

4 Historical joint-incidence surplus

4.1 Definition and exact matrix form

For a query S, define

$$o_b(S) = \sum_{s \in S} B_{bs} = |b \cap S|, \qquad u_b(S) = \sum_{s \in S} a_s B_{bs}. \tag{7}$$

The pure surplus is

$$\Delta_{\mathrm{inc}}(c \mid S) = \beta \sum_b B_{bc} v_b u_b(S) \max\{0, o_b(S) - 1\}. \tag{8}$$

For a batch, with $\odot$ denoting elementwise multiplication,

$$D_{\mathrm{inc}} = \beta B^\top \left[(D_v B D_a J) \odot (BJ - \mathbf{1})_+ \right]. \tag{9}$$

`score_simplicial_incidence_only` returns this array and its propagated version. It does not add (3).

Proposition 4.1 (Nonnegativity and exact activation)**.** *Assume $a_s > 0$, $v_b > 0$, and $\beta > 0$. For $c \notin S$, $\Delta_{\mathrm{inc}}(c \mid S) > 0$ if and only if a training bundle contains c and at least two distinct seeds. In particular, $|S| \le 1$ implies $\Delta_{\mathrm{inc}} = 0$. With $\beta = 0$, the surplus is identically zero.*

Proof. Each summand in (8) is nonnegative. It is strictly positive exactly when $B_{bc} = 1$ and $o_b \ge 2$, since these conditions also imply $u_b > 0$. A finite nonnegative sum is positive exactly when one summand is. □

4.2 What order of interaction is actually represented?

Proposition 4.2 (Triple-incidence representation)**.** *Let $T_{cst} = \sum_b v_b B_{bc} B_{bs} B_{bt}$ for distinct c, s, t. For candidates outside S,*

$$\Delta_{\mathrm{inc}}(c \mid S) = \beta \sum_{\{s,t\} \subset S} (a_s + a_t) T_{cst}. \tag{10}$$

Thus the direct surplus is a query-quadratic aggregation of weighted three-item incidences, even when training hyperedges contain more than three items.

Proof. For a particular bundle, write $x_s = B_{bs} \in \{0, 1\}$. Since $x_s^2 = x_s$,

$$u_b(o_b - 1) = \sum_s a_s x_s \left(\sum_t x_t - 1 \right) \tag{11}$$

$$= \sum_{s \ne t} a_s x_s x_t = \sum_{\{s,t\} \subset S} (a_s + a_t) x_s x_t. \tag{12}$$

If $o_b = 0$, both u_b and the expression are zero; if $o_b \geq 1$, the clamp in (8) does not change the product. Substituting this identity into (8) gives (10). □

This distinction matters. The operator accesses genuinely joint information absent from a pairwise graph, but it is not an unrestricted estimator of all k-way interactions for every k. Two hypergraphs with matching T and matching normalization statistics produce the same direct surplus. There is no boundary operator, orientation, Hodge Laplacian, or homology calculation in this function. "Simplicial" names its joint-incidence construction; it does not establish a topological learning guarantee.

4.3 Separation from identical pairwise projections

Consider the following unweighted hypergraphs on items $0, \ldots, 5$:

$$\mathcal{H}_1 = \{012, 034, 135, 245\}, \tag{13}$$

$$\mathcal{H}_2 = \{013, 024, 125, 345\}. \tag{14}$$

Here 012 denotes the set $\{0, 1, 2\}$. Each item has degree two, all bundles have size three, and the off-diagonal pairs in both systems are

$$\{01, 02, 12, 03, 04, 34, 13, 15, 35, 24, 25, 45\}.$$

Consequently, the full $B^\top B$, including its diagonal, agrees between the systems. So do W, P_{sym}, and P_{rw}.

Theorem 4.3 (Pairwise information is insufficient)**.** *No function of the common pairwise projection and its shared degree/size statistics can reproduce* (8) *for every hypergraph.*

Proof. For $S = \{0, 1\}$ and $c = 2$, $\mathcal{H}_1$ has the contributing bundle 012, giving $\Delta_{\text{inc}} = \beta v_{012}(a_0 + a_1) > 0$. In $\mathcal{H}_2$, no bundle contains $0, 1, 2$, so $\Delta_{\text{inc}} = 0$. A deterministic function of the identical pairwise inputs must give the same output in both cases and therefore cannot equal both surplus values. □

This is an expressivity separation, not an identification theorem for a whole hypergraph, and not a statistical advantage on every dataset. The matched sizes avoid confounding the comparison with volume weights.

4.4 Activation rate, propagation, and retrieval

Define the query event

$$E(S) = \{\exists b \in \mathcal{B}_{\text{tr}} : |b \cap S| \geq 2\}. \tag{15}$$

For a random query, let $A(S)$ mean that at least one candidate outside S has a nonzero direct surplus. Then

$$\Pr(A(S)) \leq \Pr(E(S)). \tag{16}$$

The inequality can be strict: an overlapping bundle may contain seeds only. If $E(S)$ is false, the entire surplus vector is zero and linear propagation also returns zero. If it is true, propagation can deliver signal to candidates that do not themselves share the original hyperedge.

Corollary 4.4 (An opportunity bound on a bounded metric)**.** *Suppose two scoring pipelines differ only by this surplus and its deterministic propagation, with identical remaining parameters and tie handling. For a per-query metric $M \in [0, 1]$,*

$$|\mathbb{E}[M_{\text{aug}} - M_{\text{base}}]| \leq \Pr(E(S)). \tag{17}$$

Proof. Outside $E(S)$ the two scores and rankings agree, so the difference is zero. Inside the event the metric difference has absolute value at most one. Apply the triangle inequality and take expectations. □

This loose bound applies only to a controlled comparison with everything else fixed. Retuning global coefficients after adding the channel can change inactive-query scores and invalidates that argument. The bound does not determine the sign of improvement on active queries.

5 Quadratic concordance from candidate–seed supports

5.1 Nonnegative support construction

For each candidate and seed, the Rao operator uses

$$w_{cs} = W_{cs} + \eta \max\{0, x_c^\top x_s\}, \qquad \eta \geq 0, \tag{18}$$

when dense features are supplied. Otherwise the second term is absent. Theoretical cosine bounds below assume unit-norm feature rows; zero rows may be retained. The function does not normalize arbitrary inputs itself. Write $A_c = \sum_{s \in S} w_{cs}$. If $A_c > 0$, set $p_{cs} = w_{cs}/A_c$; when $A_c = 0$, define all concordance scores to be zero.

5.2 Elementary symmetric polynomial identity

Let $e_2(w_c) = \sum_{s<t} w_{cs} w_{ct}$. The unweighted operator is

$$R_0(c \mid S; \tau) = \frac{2e_2(w_c)}{A_c + \tau} = \frac{A_c^2 - \sum_s w_{cs}^2}{A_c + \tau}. \tag{19}$$

The standalone `score_concordance_e2` function uses $w_{cs} = W_{cs}$ without the dense addition. Equation (19) also describes the unweighted kernel case of the general Rao formula for any supplied nonnegative supports, including (18).

Proposition 5.1 (Mass–diversity factorization)**.** *For $A_c > 0$,*

$$R_0 = \frac{A_c^2}{A_c + \tau} \left(1 - \sum_s p_{cs}^2\right). \tag{20}$$

If exactly $k_+ > 0$ seeds have positive support, then

$$0 \leq R_0 \leq \frac{A_c^2}{A_c + \tau} \left(1 - \frac{1}{k_+}\right), \tag{21}$$

with equality at the upper bound for equal support on those k_+ seeds.

Proof. Expanding $(\sum_s w_{cs})^2$ yields the sum of squares plus $2e_2$. Substitution of $w_{cs} = A_c p_{cs}$ proves (20). Cauchy–Schwarz gives $1 = (\sum p_{cs})^2 \leq k_+ \sum p_{cs}^2$. The lower bound follows from nonnegative cross-products, and equal shares attain the upper bound. □

At fixed total mass, concordance rewards dispersed support. Without fixing mass, it trades diversity against magnitude. The code clamps a small negative value of $A_c^2 - \sum_s w_{cs}^2$ to zero to guard cancellation.

5.3 Relation to products and its exact limit

For $m_S = |S|$,

$$\prod_{s \in S} (\tau + w_{cs}) = \sum_{j=0}^{m_S} \tau^{m_S - j} e_j(w_c), \qquad e_0 = 1. \tag{22}$$

This identity explains how e_2 occurs as one term in a product expansion. It does not make R_0 equal to a product-of-experts likelihood: other terms are omitted and the denominator is different. Strict conjunction would depend on all seeds, for example through $e_{m_S} = \prod_s w_{cs}$. For equal total mass three,

$$\begin{aligned} &w^{(1)} = (1.5, 1.5, 0), && e_2(w^{(1)}) = 2.25, && e_3(w^{(1)}) = 0, \\ &w^{(2)} = (2.1, 0.45, 0.45), && e_2(w^{(2)}) = 2.0925, && e_3(w^{(2)}) = 0.42525. \end{aligned}$$

Concordance prefers the first, while the strict product prefers the second. The use of second-order terms resembles established bilinear models such as factorization machines [3]; the support normalization and seed kernel specify the particular score used here.

6 Redundancy weighting through Rao diversity

6.1 Similarity and weighting matrices

With category labels ℓ_s and normalized features, the default combined seed similarity is

$$G_{st} = \tfrac{1}{2}\mathbf{1}\{\ell_s = \ell_t\} + \tfrac{1}{2}\max\{0, x_s^\top x_t\}, \quad s \neq t, \qquad G_{ss} = 1. \tag{23}$$

With only one source, that source determines G. With neither, the code uses $G = I$. For $s \neq t$, its general kernel is

$$K_{st}^{(\gamma)} = \begin{cases} \max\{0, 1 + \gamma G_{st}\}, & \gamma < 0, \\ 1 + \gamma G_{st}, & \gamma \geq 0, \end{cases} \qquad K_{ss}^{(\gamma)} = 0. \tag{24}$$

Complementarity mode has $\gamma = -1$, giving $K_{st} = 1 - G_{st}$. The music evaluator instead uses $\gamma = 1$, which favors agreement between similar seeds. That is a coherence weighting, not the same redundancy penalty. In either case, missing or inaccurate metadata changes what "different" or "similar" means.

6.2 Quadratic diversity and implementation

The score is

$$R_K(c \mid S; \tau) = \frac{w_c^\top K_S w_c}{A_c + \tau} = \frac{A_c^2}{A_c + \tau}\, p_c^\top K_S p_c. \tag{25}$$

For a dissimilarity matrix K, $p^\top K p$ is Rao's quadratic diversity [2]. It is also the expected dissimilarity between two independent draws of a seed index from p. The independence here is a mathematical device for interpreting the double sum; it does not assert independence of the observed seed evidence. The sum counts both orders of each unordered seed pair, consistently with $2e_2$ in (19).

Proposition 6.1 (Bounds and vanishing conditions)**.** *If K is symmetric, zero-diagonal, and $0 \leq K_{st} \leq K_{\max}$, then*

$$0 \leq R_K \leq K_{\max} \frac{A_c^2}{A_c + \tau} \left(1 - \sum_s p_{cs}^2\right) \leq K_{\max} A_c. \tag{26}$$

The score is zero if every pair with positive support has $K_{st} = 0$. It is positive exactly when at least one pair has positive support on both seeds and a strictly positive kernel entry.

Proof. Expand $w^\top K w = \sum_{s \neq t} K_{st} w_s w_t$. All terms are nonnegative. Bounding each kernel entry by $K_{\max}$ and applying (20) gives (26). The positivity condition follows from the same finite nonnegative sum. □

Complementarity mode has $K_{\max} = 1$. For normalized features and $\gamma = 1$, $K_{\max} = 2$. These bounds concern raw scores, before catalogue z-normalization. Neither positive semidefiniteness nor a metric triangle inequality is needed for these bounds.

6.3 Redundant and diverse seeds

If all seeds have $G_{st} = 1$, complementarity-mode concordance is zero. If all off-diagonal similarities are zero, $R_K = R_0$. This formalizes the difference between redundant and distinct support sources, conditional on the metadata proxy. Different categories are not by themselves proof of functional compatibility; dissimilar items can also be incompatible.

The operator is invariant to permutation of the same seed set when the corresponding rows and columns of K are permuted together. It is not generally invariant to adding a distinct item identifier with identical metadata to an existing seed. Such an addition changes A_c, the support distribution, and its cross-terms with other seed groups. Exact duplication-invariance would require a separately specified group-level aggregation; it is not a property claimed here.

Nor is R_K coordinatewise monotone for every admissible K. For example, take three seeds with $K_{12} = K_{21} = 1$ and all other entries zero. At $w = (1, 1, t)$,

$$R_K(w; \tau) = \frac{2}{2 + t + \tau}, \tag{27}$$

which decreases with t. A new support source that contributes no dissimilarity-weighted agreement can dilute the normalized score.

6.4 Scaling law and exact finite-scale correction

For a fixed seed kernel and any $\lambda > 0$,

$$R_K(\lambda w; \lambda \tau) = \lambda R_K(w; \tau). \tag{28}$$

This follows because the numerator scales quadratically and the denominator linearly. For fixed τ, writing $Q_K = w^\top K w$ instead yields

$$R_K(\lambda w; \tau) = \frac{\lambda^2 Q_K}{\lambda A + \tau}. \tag{29}$$

For $Q_K > 0$, the ratio to $\lambda R_K(w; \tau)$ is

$$\frac{R_K(\lambda w; \tau)}{\lambda R_K(w; \tau)} = \frac{\lambda(A + \tau)}{\lambda A + \tau}. \tag{30}$$

Thus exact degree-one homogeneity requires joint scaling. The large-support approximation replaces $A + \tau$ by A; its relative shortfall from Q_K/A is exactly $\tau/(A + \tau)$. Merely letting λ grow with fixed original A does not prove equality to $\lambda R_K(w; \tau)$. If support is rescaled by changing underlying model parameters, the dense and graph terms in (18) must both scale to invoke (28).

6.5 Information separation from incidence

Rao concordance is a function of candidate–seed supports and seed similarities. If those inputs agree, its output agrees. In particular, the two hypergraphs in (14), with matched metadata, yield identical concordance but different incidence surplus. Conversely, concordance can be positive when its supports come from distinct pairwise bundles and no single bundle contains two seeds. The channels therefore access different information. This does not imply statistical independence, uncorrelated outputs, or orthogonality of fitted estimators.

7 Fusion, metadata priors, and interpretation of the target

7.1 Catalogue normalization and fused observations

For a channel $g_k(c, S)$, define column-wise catalogue normalization

$$z_k(c, S) = \frac{g_k(c, S) - \bar{g}_k(S)}{s_k(S) + \varepsilon_z}, \qquad \bar{g}_k(S) = \frac{1}{n} \sum_i g_k(i, S). \tag{31}$$

The repository uses $\varepsilon_z = 10^{-9}$. NumPy uses its default population standard deviation, while the Torch implementation uses its default sample correction. This small convention difference is part of the implementation, not a new statistical estimator. Typically,

$$Y_{cS} = \sum_k \omega_k z_k(c, S), \qquad P_{cS} = \sum_{k \in \mathcal{P}} \xi_k z_k(c, S), \qquad \mu_{cS} = s_{\text{val}} P_{cS}. \tag{32}$$

The selected $\mathcal{P}$ need not contain only exogenous metadata: the controlled retail prior includes bridge, text, auxiliary interaction, and Rao terms. Y and P can therefore share channels and training data. The term "prior" describes its role in shrinkage, not a guarantee that it is independent evidence. The moment model below must be assessed for the actual fused scores, including this dependence.

7.2 The latent target is a model, not an observed label

Let $\theta_{cS} \in \mathbb{R}$ denote latent compatibility on the standardized score scale. It is not defined here as a probability, a calibrated log-odds, or a statistically identified log-lift. The score Y is a constructed collaborative/content estimator, not a direct repeated measurement of an observed θ. The Gaussian hierarchy supplies a working model connecting the two. Algebraically exact posterior updating under that model is a narrower statement than correct real-world calibration of θ.

8 Exposure-modulated Gaussian posterior

8.1 Assumptions

For fixed S, c, positive exposure f_c, and a fitted prior mean μ_{cS}, assume

$$\theta_{cS} \mid \mu_{cS}, f_c \sim \mathcal{N}(\mu_{cS}, \sigma_0^2), \tag{33}$$

$$Y_{cS} \mid \theta_{cS}, \mu_{cS}, f_c \sim \mathcal{N}(\theta_{cS}, \sigma_\epsilon^2 / f_c), \tag{34}$$

with strictly positive variances. The second line specifies a zero-mean likelihood error conditional on the latent compatibility and prior mean. It does not follow merely from positive co-occurrence counts. Frequency is treated as an exposure proxy, although real histories can have unequal opportunity, temporal drift, and dependent observations.

8.2 Derivation by completing the square

Suppress the indices and put $v_f = \sigma_\epsilon^2 / f$. The posterior log-density, up to constants independent of θ, is

$$\log p(\theta \mid Y, \mu, f) = -\frac{1}{2} \left[\frac{(\theta - \mu)^2}{\sigma_0^2} + \frac{(Y - \theta)^2}{v_f} \right] + C \tag{35}$$

$$= -\frac{1}{2} \left[(\sigma_0^{-2} + v_f^{-1})\theta^2 - 2(\mu\sigma_0^{-2} + Y v_f^{-1})\theta \right] + C'. \tag{36}$$

Therefore the posterior is Gaussian with

$$V_f = \left(\frac{1}{\sigma_0^2} + \frac{f}{\sigma_\epsilon^2}\right)^{-1}, \tag{37}$$

$$\widehat{\theta} = V_f\left(\frac{\mu}{\sigma_0^2} + \frac{fY}{\sigma_\epsilon^2}\right) = \frac{fY + \kappa\mu}{f+\kappa} = (1-u_f)Y + u_f\mu, \quad u_f = \frac{\kappa}{f+\kappa}, \quad \kappa = \frac{\sigma_\epsilon^2}{\sigma_0^2}. \tag{38}$$

This is the literal update in `conjugate_posterior` and `apply_eb_shrinkage`; no positive-part residual is added.

Proposition 8.1 (Convexity, exposure limits, and sensitivity)**.** *For $f \geq 0$ and $\kappa > 0$, $0 < u_f \leq 1$ and the posterior mean lies between Y and μ. Moreover,*

$$\frac{\partial u_f}{\partial f} = -\frac{\kappa}{(f+\kappa)^2}, \qquad \frac{\partial \widehat{\theta}}{\partial Y} = \frac{f}{f+\kappa}, \qquad \frac{\partial \widehat{\theta}}{\partial \mu} = \frac{\kappa}{f+\kappa}. \tag{39}$$

At $f = 0$ the zero-likelihood-precision limit is $\widehat{\theta} = \mu$; as $f \to \infty$, $\widehat{\theta} \to Y$ for fixed Y, μ, κ.

The $f = 0$ case is a limit and a software convention; the positive-exposure Gaussian likelihood in (34) is not a proper finite-variance observation distribution at $f = 0$. In contrast, `kappa_f=0` explicitly disables shrinkage and returns Y even on cold items. It should not be interpreted by substituting $f = \kappa = 0$ in the fraction.

For $f = (0, 0, 9)$, $Y = (-2, -2, 4)$, $\mu = (-1, 1, 1)$ and $\kappa = 4$,

$$\widehat{\theta} = (-1, 1, 40/13) \approx (-1, 1, 3.076923). \tag{40}$$

Coldness alone does not impose a positive score: a negative cold prior remains negative. Whether that score enters a top-K list still depends on every competing candidate.

8.3 A conditional squared-error guarantee, with its limits

Theorem 8.2 (Bayes risk under the specified hierarchy)**.** *For known variances and a correctly specified hierarchy, the posterior mean minimizes conditional expected squared error among all measurable estimators based on Y, μ, f. Its integrated risk for fixed $f > 0, \mu$ is*

$$\mathbb{E}[(\widehat{\theta} - \theta)^2 \mid f, \mu] = \frac{\sigma_0^2\sigma_\epsilon^2}{f\sigma_0^2 + \sigma_\epsilon^2} \leq \min\{\sigma_0^2, \sigma_\epsilon^2/f\}. \tag{41}$$

Proof. For any $a = a(Y, \mu, f)$, the conditional squared-error decomposition is

$$\mathbb{E}[(\theta - a)^2 \mid Y, \mu, f] = \mathrm{Var}(\theta \mid Y, \mu, f) + (a - \mathbb{E}[\theta \mid Y, \mu, f])^2. \tag{42}$$

The second term is minimized at the posterior mean. The posterior variance is the constant V_f in (37), so integrating over Y gives (41). Its denominator is at least either individual precision contribution, giving both inequalities. □

This is an established property of posterior means, not a new SIMBA optimality theorem. Estimated or validation-selected hyperparameters, biased Y, prior misspecification, and correlated prior/observation errors can invalidate the comparison. Lower latent squared error also does not imply better retrieval of held-out binary memberships. The implementation returns a plug-in mean; it does not return uncertainty intervals accounting for fitted hyperparameters or a verified coverage guarantee.

9 Variance-component estimation and identifiability

9.1 From the hierarchy to a conditional moment

Write $\theta - \mu = \zeta$ and $Y - \theta = \epsilon$, so $Y - \mu = \zeta + \epsilon$. Under (34),

$$\mathbb{E}[(Y-\mu)^2 \mid f, \mu] = \sigma_0^2 + \sigma_\epsilon^2/f. \tag{43}$$

The cross-term vanishes because $\mathbb{E}[\epsilon \mid \zeta, f, \mu] = 0$. Gaussianity is sufficient for the posterior derivation, but not necessary for this moment: zero means, the stated variances, and a zero cross-moment are enough. Conversely, the moment alone does not imply a Gaussian posterior.

With Q_v validation queries, the implemented item-level response is

$$v_c = \frac{1}{Q_v} \sum_{S \in \mathcal{Q}_v} (Y_{cS} - \mu_{cS})^2, \qquad x_c = 1/f_c, \quad c \in \mathcal{W} = \{c : f_c > 0\}. \tag{44}$$

The method regresses v_c on x_c; it does not regress the variance of Y alone, nor the spread of the prior means. Each warm item receives one OLS observation regardless of its exposure.

9.2 Raw OLS formulas and their proof conditions

Let $N_w = |\mathcal{W}|$, $D_x = \sum_{c \in \mathcal{W}} (x_c - \bar{x})^2$. When $N_w \geq 2$ and $D_x > 0$, the raw estimates are

$$\widetilde{\sigma}_\epsilon^2 = \frac{\sum_c (x_c - \bar{x})(v_c - \bar{v})}{D_x}, \tag{45}$$

$$\widetilde{\sigma}_0^2 = \bar{v} - \widetilde{\sigma}_\epsilon^2 \bar{x}. \tag{46}$$

They solve the unweighted least-squares normal equations for an intercept and slope. Calling these values variance components describes their intended target; finite-sample values may be negative.

Proposition 9.1 (Conditional unbiasedness of raw components)**.** *For a fixed identifiable design vector x, suppose $v_c = \sigma_0^2 + \sigma_\epsilon^2 x_c + e_c$ and $\mathbb{E}[e_c \mid x] = 0$ for each warm item. Then the raw slope and intercept in (46) are conditionally unbiased.*

Proof. Substitution into the slope gives

$$\widetilde{\sigma}_\epsilon^2 = \sigma_\epsilon^2 + \frac{\sum_c (x_c - \bar{x}) e_c}{D_x}. \tag{47}$$

Conditional expectation of the second term is zero. The intercept error is $\bar{e} - \bar{x}(\widetilde{\sigma}_\epsilon^2 - \sigma_\epsilon^2)$, which also has conditional expectation zero. □

Independence across items and homoskedastic errors are not needed for this expectation calculation. They do matter for usual OLS standard errors, efficiency, and concentration. Shared queries and graph structure can correlate residuals, so treating $N_w Q_v$ cells as independent observations would be unjustified. The routine does not estimate regression standard errors.

There is an additional fitted-score caveat: Y and μ are constructed and their scale or coefficients can be estimated on these same validation queries. The conditional-mean-zero premise is not automatically preserved after that fitting and selection. The raw formulas are OLS; unconditional finite-sample unbiasedness of the complete fitted SIMBA pipeline has not been established. A separate calibration split or cross-fitting could make such an analysis cleaner, but is not implemented by this document.

9.3 Identifiability and the numerical decision rule

Proposition 9.2 (Moment identifiability)**.** *The intercept and slope are separately identifiable from the linear moment equation exactly when the design matrix with columns* $1, x$ *has rank two, equivalently when at least two distinct warm inverse frequencies are observed. With a single common* x_0*, only* $\sigma_0^2 + x_0\sigma_\epsilon^2$ *is identified.*

Proof. The normal matrix has determinant

$$\det \begin{pmatrix} N_w & \sum x_c \\ \sum x_c & \sum x_c^2 \end{pmatrix} = N_w D_x. \tag{48}$$

It is invertible exactly when $D_x > 0$. For constant x_0, every pair of components with the same intercept-plus-x_0-times-slope has the same fitted conditional moment. □

The code uses the stronger numerical requirement $N_w \geq 2$ and $D_x > 10^{-12}$, testing a *sum* of squared centered inverse frequencies, not a sample variance threshold. It otherwise returns `identifiable=False`, null raw and constrained components, and `kappa_f=0.0`. All production evaluator grids then restrict their candidate shrinkage values to zero. Non-identification is not an estimate that the true noise component equals zero; disabling shrinkage is the chosen fallback policy.

9.4 Positive truncation and ratio clipping

For identifiable designs, the returned plug-in components are

$$\widehat{\sigma}_0^2 = \max\{\widetilde{\sigma}_0^2, 10^{-4}\}, \qquad \widehat{\sigma}_\epsilon^2 = \max\{\widetilde{\sigma}_\epsilon^2, 10^{-4}\}, \tag{49}$$

$$\widehat{\kappa} = \mathrm{clip}(\widehat{\sigma}_\epsilon^2/\widehat{\sigma}_0^2, 0.05, 50). \tag{50}$$

The raw outputs are unrounded; constrained outputs are reported separately. The nonlinear transformations in (50) are generally biased, and even an unclipped ratio of unbiased components need not be unbiased. These componentwise floors are not the joint solution of a constrained nonnegative least-squares problem. "Constrained plug-in" refers to admissible returned values, not to an optimization problem that the code does not solve. Some evaluators subsequently choose a different positive κ from a validation grid; that is a tuned posterior-shaped score, not necessarily the direct moment-ratio estimate.

9.5 Prior-shift invariance and misspecification

For any array h_{cS}, replacing (Y, μ) by $(Y + h, \mu + h)$ leaves v_c and the variance estimates unchanged. This exact residual invariance explains why a simulated change in the spread of prior means need not change $\widehat{\kappa}$ when the same prior is added to Y. It does not prove that estimating a different prior on real data is harmless.

For example, if the residual has conditional bias $b(f)$ and prior/noise cross-covariance $q(f)$, its second moment can instead contain

$$\mathbb{E}[(Y - \mu)^2 \mid f] = \sigma_0^2 + \sigma_\epsilon^2/f + b(f)^2 + 2q(f), \tag{51}$$

under an otherwise comparable decomposition. Frequency-dependent prior error or correlated reused channels can then be mistaken for exposure noise. Diagnostics should examine residual-moment curves against $1/f$, negative raw components, clipping frequency, and sensitivity to the prior.

10 Validation-frozen inference and a batch-invariance result

10.1 Calibration contract

A typical prior scale is

$$s_{\text{val}} = \frac{\text{std}(Y_{\text{val}})}{\text{std}(P_{\text{val}}) + 10^{-9}}, \tag{52}$$

where the standard deviations pool the validation score array. Individual evaluators differ slightly in stabilizers and whether scale one is also considered. Every selected scale is frozen before test inference. The same applies to variance ratios, channel weights, encoder parameters, and any backward-elimination decisions. A moment fit uses validation scores without requiring held-out target labels; validation metric selection can use validation targets. Neither operation is performed on the test-query score batch in the current pipelines.

Theorem 10.1 (Score independence from unrelated query columns)**.** *Fix the catalogue, trained operators, metadata, and calibration parameters. Suppose each channel for query S is computed only from S and those fixed objects; any z-normalization is across catalogue items within its own query column. Then appending unrelated query columns to a batch does not change $\widehat{\theta}_{cS}$ in exact arithmetic.*

Proof. The raw channel column for S is unchanged by hypothesis. Its mean and standard deviation in (31) use that column only, so the normalized column is unchanged. Fixed linear fusion gives unchanged Y_{cS} and P_{cS}. Frozen scaling gives unchanged μ_{cS}, and (38) uses only these values and fixed f_c, κ. □

Re-estimating (52) on each test batch would violate the frozen parameter premise. Floating-point kernels may introduce tiny layout-dependent differences, and evaluator tie perturbations can affect exact tied ranks; the proposition concerns the deterministic mathematical score. It does not assert invariance to changing the catalogue, because catalogue means, standard deviations, and available candidates can then change.

11 Auxiliary channels and boundaries of the theory

11.1 Ontology and slot saturation

The ontology operator forms category co-occurrence statistics and positive-PMI-like off-diagonal weights, with optional soft clusters and within-type channels. `score_slot_saturation` adds a penalty based on query seeds sharing a candidate's category and empirical category duplication rates. Despite its name, this is an additive pairwise diagonal correction. It is not a distinct higher-order joint-incidence estimator. Its interpretation depends on the category system and on whether a bundle is expected to fill distinct slots or contain repeated types.

11.2 Spectral features and content bridges

The graph operator also constructs positive-PMI co-occurrence features and a truncated-SVD embedding. Content bridges mix frozen features with one or more graph propagation steps. Their scores supply additional channels and help score training-cold items with catalogue features. These operations do not make cold items collaboratively observed, and their feature quality is not guaranteed by the incidence or posterior proofs. A content-neighbour matrix can be row-normalized on populated rows without making every graph smoother in the system stochastic.

11.3 Set-attention encoder

The full evaluators additionally train a small neural set encoder on training examples, using transformed item features, attention pooling, and a contrastive objective with sampled negatives. Its output is fused as another channel. A set sum or attention aggregation is permutation invariant when attention weights are computed equivariantly and no order-dependent input is introduced. That structural observation is not a convergence or generalization theorem for the trained encoder. The offline demo and controlled retail ablation do not train this model; their execution does not verify the full encoder's optimization.

11.4 Pruning and retrieval policies

`prune_weights_val` performs validation-tolerance backward elimination of optional channels. It is not a minimum-description-length objective, and no global optimum over channel subsets is claimed. The optional tail/cold slot reservation is a post-hoc retrieval policy, separate from the normal–normal posterior. If enabled, the resulting ranking is not solely the ordering of (38). Reports should state whether this policy was used.

12 Controlled ablations: what is and is not fixed

In the retail ablation, let L denote the collaborative pairwise mixture, P_g the fixed dataset-specific graph propagation, and B_r, T, U the bridge, text, and auxiliary interaction scores. Writing z for catalogue normalization, the implemented ladder is

$$F_0 = z(L) + 0.8z(P_g L) + 0.8z(B_r) + 0.5z(T) + 0.5z(U), \tag{53}$$

$$F_1 = z(L + \Delta_{\text{inc}}) + 0.8z(P_g(L + \Delta_{\text{inc}})) + 0.8z(B_r) + 0.5z(T) + 0.5z(U), \tag{54}$$

$$F_2 = F_0 + 0.35z(R_K), \tag{55}$$

$$F_3 = F_1 + 0.35z(R_K), \tag{56}$$

$$F_4 = (1 - u)F_3 + u\mu. \tag{57}$$

The prior before frozen scaling is $P = 0.8z(B_r) + 0.5z(T) + 0.5z(U) + 0.35z(R_K)$. The graph, data split, raw backbone channels, and numerical coefficients are shared within a seed. Only validation scores fit the final prior scale and moment ratio.

Because $z(L + \Delta_{\text{inc}}) \neq z(L) + z(\Delta_{\text{inc}})$ in general, adding incidence also changes the normalization of the combined collaborative channel. The comparison measures the implemented incidence-plus-normalization intervention, not a fixed standardized baseline plus a raw surplus. Likewise $F_4 - F_3 = u(\mu - F_3)$ both attenuates and adds evidence; it is not a pure cold-item bonus. The $F_1 - F_0$ and $F_2 - F_0$ comparisons isolate the two additions, while F_3 checks their combination and F_4 adds posterior calibration. This is a controlled ladder with two individual mechanism branches, not a complete factorial study of all interactions.

For a held-out set H, Recall@K is

$$\text{Recall@}\,K(S, H) = \frac{|\,\text{TopK}(\mathcal{I} \setminus S) \cap H|}{|H|}. \tag{58}$$

Split-seed means and sample standard deviations describe split variation. A positive mean difference alone does not establish significance or causality. Paired per-query or per-split comparisons, exposure strata, activation-conditioned metrics, and explicit tie handling strengthen interpretation. Current evidence and its provenance remain in `docs/RESULTS.md`; this theory document adds no new benchmark results.

13 Computation, memory, and weighted-input caveats

13.1 Operator costs

Let $L_B = \mathrm{nnz}(B)$, $L_W = \mathrm{nnz}(W)$, and $k = |S|$. Factored incidence for Q dense output columns takes on the order of $L_B Q$ sparse–dense work plus bundle/query intermediates and an nQ output. It avoids storing an n^3 triple tensor even though (10) characterizes its information. Computing graph diffusion in factored form also uses order $L_B Q$ work.

Unweighted concordance can use WJ and $W^{\odot 2}J$, with sparse products rather than explicit enumeration of seed pairs. In the general Rao implementation, each query materializes an $n \times k$ support array and multiplies it by a $k \times k$ kernel: the dense quadratic work is $O(nk^2)$, with support memory $O(nk)$, plus output storage and any dense feature products. A dense feature width d adds $O(ndk)$ work for candidate–seed dot products. Full-catalogue score banks can therefore require several GiB even when a neural encoder has few parameters.

Preprocessing costs also matter: forming $B^\top B$ can generate up to order $\sum_b d_b^2$ pair contributions before aggregation. Sparse input does not imply that its projection remains sparse. SVD, auxiliary graphs, clustering, and model training add costs beyond the scoring formulas. No sublinear full-catalogue retrieval guarantee is claimed.

13.2 Optional weighted constructor

The constructor accepts bundle weights. Let H be binary incidence, $t_b \geq 0$ the supplied weight, and $\widetilde{B} = D_t H$ the stored nonbinary matrix. Then exposure is $f_i = \sum_b t_b H_{bi}$, bundle sizes come from H, and a bilinear projection uses

$$\widetilde{B}^\top D_v \widetilde{B} = H^\top \operatorname{diag}(t_b^2 v_b) H. \tag{59}$$

Weights are squared because they enter both factors. The surplus also contains t_b^2, while overlap counts use H. It is not a once-weighted hyperedge sum unless all weights are one or their interpretation is adjusted accordingly.

There is a further binary-specific diagonal correction in `propagate`. For weighted input the implemented matrix is

$$\widetilde{P} = D_a \widetilde{B}^\top D_r \widetilde{B} D_a - \operatorname{diag}\left(a^2 \odot (\widetilde{B}^\top r)\right), \tag{60}$$

whose diagonal entry is

$$\widetilde{P}_{ii} = a_i^2 \sum_b r_b (t_b^2 - t_b) H_{bi}. \tag{61}$$

Thus the zero-diagonal and nonnegative-smoother statements for binary B do not automatically extend to arbitrary bundle weights. The shipped evaluators instantiate the constructor without this optional weighting. These caveats specify the existing function rather than silently proving a property for a different weighted operator.

14 Verification map and remaining scientific questions

14.1 What the existing suite checks

The mathematical audit checks active-row stochasticity, pure incidence activation, separation of a joint bundle from a pairwise triangle, the unweighted concordance identity, redundancy examples, numerical posterior values, simulated variance-component recovery, equal-frequency non-identification, frozen-scale batch invariance, validation pruning, and production evaluator compliance with the non-identification policy. The standalone suite checks packaging, data utilities, the offline demo, and fail-closed evaluation orchestration.

Unit tests instantiate examples and catch implementation regressions; they do not prove all quantified statements in this document. In particular, the fully degree-matched pairwise

separation example in (14) is justified by the explicit enumeration above, rather than by assuming the simpler triangle test matches every diagonal statistic.

14.2 What remains empirical or unresolved

The following are scientific questions rather than consequences of the preceding algebra:

1. Does training exposure approximate inverse observation variance after normalization and feature fusion? Is $1/f$ better than alternatives?
2. Does the metadata-derived K distinguish redundant from complementary constraints in the target application?
3. Does incidence help on active queries, especially after matching popularity, query size, and candidate frequency?
4. Are posterior gains robust to prior quality, channel reuse, split selection, and variance clipping? How often are raw components negative or numerically unidentified?
5. Do gains persist against strong tuned methods under the same candidate catalogue, data access, split, and computational budget?
6. Can a meaningful generative target or retrieval guarantee be established beyond a working standardized-score hierarchy?

The present algorithm combines distinct mechanisms and exposes testable predictions about their behavior. Its mathematical guarantees are those proved above, under their stated domains. Novelty of the full method and its publication value additionally require related-work comparison and empirical evidence; renaming established estimators does not supply that evidence.

A Reference inference procedure

1. Construct train-only incidence, degrees, collaborative graphs, and any allowed auxiliary interaction channels. Prepare the fixed catalogue features under the declared transductive or inductive protocol.
2. For validation queries, compute pairwise, incidence, concordance, content, and optional learned scores. Normalize each query column across the fixed catalogue; construct Y_{val} and P_{val}.
3. Fit the validation prior scale, compute residual moments, and return both raw and admissible variance components. If unidentified, restrict the shrinkage grid to zero. Complete any validation selection.
4. Freeze all fitted objects and parameters. For each test query, compute its own score column and apply the chosen posterior formula. Do not refit scales or variance components on test-input batches.
5. Exclude seeds, apply the declared tie convention and any explicitly enabled post-hoc policy, then select the top K eligible candidates.
6. Compute metrics from held-out test targets only after scoring. Record split seeds, data paths/provenance, parameters, and policy settings.

B Source map and build instructions

The authoritative detailed source is `docs/THEORY.tex`. A navigation page is retained at `docs/THEORY.md`; a compiled copy is supplied as `docs/THEORY.pdf`. From the repository root, build with a standard LaTeX installation:

```
mkdir -p artifacts/theory
for pass in 1 2 3; do
  pdflatex -halt-on-error -interaction=nonstopmode \
    -output-directory=artifacts/theory docs/THEORY.tex || exit 1
done
```

Repeated passes resolve the table of contents and internal references. No BibTeX step or external figure is required. Generated files stay under the ignored `artifacts/` directory; copy the final PDF to `docs/THEORY.pdf` when refreshing its checked-in rendering. Core implementation locations are:

File	Role
`simba/operators.py`	Shared graph, incidence, concordance, ontology, and shrinkage functions
`simba/ablation_study.py`	Controlled retail scoring ladder and validation-frozen moment calibration
`simba/*_eval.py`	Dataset-specific feature banks, learned channels, validation selection, and retrieval
`tests/test_audit.py`	Mathematical and evaluator-policy examples
`tests/test_standalone.py`	Installation/data/driver checks

References

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